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A model of macroalgal decomposition

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ABSTRACT

Macroalgae are currently the focus of blue carbon, biological carbon dioxide (CO_2) removal in the ocean. Carbon exported as seaweed detritus is equivalent to ~7.5% of anthropogenic CO_2 emissions. Macroalgal blue carbon hinges on the fate of this detritus, so a reliable model of its decomposition is key. However, macroalgal decomposition has been neither consistently nor reliably modelled. Here I present a model of macroalgal decomposition. Informed by the theory of detrital photosynthesis, it treats detritus as an entity in limbo that can initially either grow or decompose. This is achieved by replacing the rate constant of the conventional exponential decay model with a logistic function of time, mirroring the sigmoidal decay of detrital photosynthesis. The resulting mathematical model has three parameters that describe the initial exponential growth rate, the photosynthetic half-life, and the final exponential decay rate once physiology is factored out. The third parameter is equivalent to the rate constant of the conventional model for dead detritus. Using Bayesian data analysis in R and Stan, I show that the statistical implementation of the mathematical model provides meaningful inference given empirical data from a variety of ecological contexts and routinely outperforms the conventional model. The presented model can describe any macroalgal decomposition trajectory, which enables formal comparison across studies for the first time. I anticipate that this will prove instrumental in resolving the debate around macroalgal CO_2 removal.

1. Introduction

In a seminal paper on mathematical ecology, Olson (1963) popularised the ordinary differential equation of leaf litter decomposition

$$\frac{dm}{dt} = -km \tag{1}$$

which describes the rate of change in litter mass m proportional to itself, a relationship that is parameterised by the decay constant k. This reflects our ecological understanding that given a theoretically limitless pool of decomposers, decomposition is proportional to the detrital pool. The equation integrates to

$$\ln\frac{m(t)}{m_0} = -kt \tag{2}$$

and simplifies to the familiar exponential decay function

$$m(t) = m_0 e^{-kt} \tag{3}$$

where $m(t)$ is a function of the initial litter mass m_0, the decay constant k and time t. Often the half-life $t_{½} = \frac{\ln 2}{k}$ (Olson 1963) is reported as a more intuitive measure of decay in terms of units of t. This function has prevailed as the conventional model of decomposition.

Interest in decomposition has been rekindled due to the hope that plants can mitigate anthropogenic climate change by fixing carbon dioxide (CO_2), assimilating it into organic carbon, and sequestering it in their biomass. Carbon sequestration timescales of decades to centuries are being considered the minimum and ten millennia or longer are preferred (Babiker et al., 2022). Plant biomass can be split into two distinct carbon sinks: living plants and the detrital pool, of which the latter is considered the more promising. Often surprising to plant ecologists, plant detritus amounts to a biomass that is generally comparable and often far greater than that of living plants (Cebrián and Duarte, 1995; Wright and Kregting, 2023). But more importantly, living plants, even when considering stable or increasing populations, are limited by biomass-density constraints (Creed et al., 2019) and lifespans of typically less than a millennium (Thomas, 2013). Detritus, on the other hand, is believed to have the potential of cumulative sequestration over much longer timescales. Fossil fuels were created via the Carboniferous detrital pathway, mostly of scale trees (Hibbett et al., 2016), and the

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hope of biological CO_2 removal essentially boils down to re-enacting fossil fuel generation.

Biological CO_2 removal in the ocean is referred to by the shorthand blue carbon (Babiker et al., 2022). Macroalgae have recently moved into the focus of blue carbon (Pessarrodona et al., 2023), primarily due to their substantial global area (Duarte et al., 2022), biomass packing (Creed et al., 2019), productivity (Pessarrodona et al., 2022) and detrital export (Pessarrodona et al., 2023; Filbee-Dexter et al., 2024). Notably, decomposition is disproportionately consequential for macroalgal blue carbon. Unlike their fellow blue carbon ecosystems—seagrass meadows, mangrove forests and salt marshes—seaweed forests cannot accumulate carbon in below-ground biomass and soil. There is thus a set limit for the living seaweed carbon sink which can be estimated as the product of 6.1 million km^2 global cover (Duarte et al., 2022), 1.2 ± 3 kg m^{-2} (mean $\pm$ standard deviation) standing biomass (Creed et al., 2019) and 28 ± 4.5 % carbon content (Pessarrodona et al., 2022). This yields 0.34 ± 0.87 kg C m^{-2} and 2.1 ± 5.3 Gt C globally, or only 19 ± 49 % of annual anthropogenic CO_2 emissions (Friedlingstein et al., 2025). However, seaweeds are thought to assimilate 0.32 ± 0.54 kg C m^{-2} y^{-1} (Pessarrodona et al., 2022) and 1.3 Gt C y^{-1} globally (Duarte et al., 2022) of which 61 ± 4.2 % is exported as detritus (Pessarrodona et al., 2023). Hence 0.19 ± 0.33 kg C m^{-2} y^{-1} and 0.81 ± 0.055 Gt C y^{-1} globally, or 7.5 ± 0.51 % of annual anthropogenic CO_2 emissions, may enter the detrital pool yearly. Note that these global standard deviations are underestimates because Duarte et al. (2022) report no uncertainty. Viewed cumulatively, this is clearly the more substantial carbon sink. Decomposition alone therefore determines whether macroalgal blue carbon is significant or not, as is illustrated by the central role it takes in the latest macroalgal blue carbon models (Filbee-Dexter et al., 2024). Despite this, macroalgal decomposition remains comparatively poorly understood.

Phycologists have never been able to properly model macroalgal decomposition. Decomposition trajectories rarely match Eq. 3 reasonably well (e.g. de Bettignies et al., 2020, Pedersen et al., 2021). Most studies have therefore either refrained from fitting a parametric model (e.g. Birch et al., 1983, Vandendriessche et al., 2007, Hamersley et al., 2015, Frontier et al., 2021) or most commonly fit linear models (e.g. Bedford and Moore, 1984, Brouwer 1996, Krumhansl and Scheibling, 2012, Filbee-Dexter et al., 2022, Frontier et al., 2022, Wright et al., 2022). Meta-analyses of macroalgal k (Eq. 3) are numerous (Enríquez et al., 1993; Banta et al., 2004; Cebrián and Lartigue, 2004; Filbee-Dexter et al., 2024; Kennedy and Blain, 2025; White and Norkko, 2025) but have generally assumed Eq. 3 by default and not assessed the validity of this model in the macroalgal context. It is evident that these meta-analyses are often hampered by the complexity and diversity of macroalgal decomposition trajectories (Kennedy and Blain, 2025; White and Norkko, 2025). The only serious attempts at modelling diverse trajectories have invariably invoked the logistic function alongside the exponential (Bourguès et al., 1996; Kennedy and Blain, 2025). This defeats the purpose of a model since logistic and exponential rate constants cannot be formally compared. Kennedy & Blain (2025) realised this when attempting to compare $t_{½}$ across exponential and logistic models. While $t_{½}$ for the latter is best estimated as the inflection point of the sigmoid, they ultimately resorted to forcing an exponential model to derive this parameter (Kennedy and Blain, 2025), which undermines fitting the logistic model in the first place.

We now know what causes the observed diversity in macroalgal decomposition trajectories. Early phycologists noticed the need to "prekill" macroalgal detritus (Josselyn and Mathieson, 1980; Rice and Tenore, 1981; Birch et al., 1983; Smith and Foreman, 1984; Brouwer 1996) if it was to behave according to the assumptions of the conventional model (Eq. 3). Recently it became evident that this is because excised macroalgal tissue can continue to photosynthesise for months (de Bettignies et al., 2020), which may allow macroalgal detritus to grow rather than decompose over the short term (Frontier et al., 2021; Pedersen et al., 2021; Filbee-Dexter et al., 2022; Frontier et al., 2022; Wright et al., 2022). To avoid semantic confusion, I hereinafter use "detritus" for all detached material and "detritus *sensu stricto*" for detached *and* dead material. There is now extensive investigation into what has since been termed detrital photosynthesis (Frontier et al., 2021; Wright and Foggo, 2021; Frontier et al., 2022; Wright et al., 2022; Wright and Kregting, 2023; Wright et al., 2024) and macroalgae have been found to be unique among plants in having detritus that photosynthesises to an extent that is likely to influence its decomposition (Wright 2026a). This knowledge enables the creation of a model that accurately describes macroalgal decomposition by taking into account the effect of detrital photosynthesis. It is probable that, ignorant of detrital photosynthesis, the conventional model has substantially underestimated macroalgal decomposition, leading to overestimated seaweed blue carbon. To remedy this, I here present a nonlinear model with three parameters that is tailored to the unique case of macroalgal decomposition. First I describe its mathematical form, then I implement it as a Bayesian statistical model, and finally I validate it against a variety of empirical examples, comparing its predictive performance to that of the conventional model for each example.

2. The mathematical model

…tossed and swept like dead leaves from one spot to another, never resting, never giving back their goodness to the earth, never fully dead but in some vast limbo between life and extinction, tossing and tumbling without end… — Philip Pullman (1994) *The Tin Princess*

Macroalgal detritus is essentially in limbo. It may grow or decay, but not indefinitely. The goal is to model this temporary limbo. There are several possible approaches to describing this mathematically. As mentioned above, the logistic function has been the only attempt so far, but its rate constant is not comparable to that of Eq. 3 and it cannot describe the full spectrum of possible decomposition trajectories (e.g. an increase followed by a decrease). Hence we must dismiss it. Another approach could be a composite function of a saturating function (e.g. rectangular hyperbola, hyperbolic tangent, exponential) and Eq. 3. This may describe all possible trajectories but is somewhat of an artificial construct and more importantly would have more parameters than are practical. The best way forward is to stick with our trusty Eq. 3 but rebuild it from its differential form (Eq. 1). Instead of treating k as constant, we can let it be a function of time:

$$\frac{dm}{dt} = k(t)m \tag{4}$$

To my knowledge, time-variant exponential decay has only been used twice in the context of decomposition, by Rovira & Rovira (2010) to describe the variety of terrestrial leaf litter decomposition trajectories and by Manojlovic (2021) to model seagrass decomposition (Trevathan-Tackett et al., 2020). Rovira & Rovira (2010) chose the form of $k(t)$ based on context while Manojlovic (2021) used a random walk, the simplest autoregression which defines k only in terms of k at the previous timepoint plus normal error:

$$k(t) \sim \text{Normal}(\ k(t - \Delta t)\ ,\ \sigma\ ) \tag{5}$$

where σ is the standard deviation of the normal distribution around the mean $k(t - \Delta t)$ and determines the k step size of the random walk. The timestep size Δt is typically set to 1. While this approach is the most flexible, it requires sophisticated Bayesian state space modelling with an ordinary differential equation solver. Manojlovic (2021) used the Bayesian language LibBi (Murray, 2015) and to my knowledge there is currently no way to implement this in the more popular Stan language (Carpenter et al., 2017) (Manojlovic, personal communication, discourse.mc-stan.org/t/random-walk-in-rstan/34308). Furthermore, being the simplest type of order 1 autoregressive process, a random walk's only parameter is σ which has no inherent directionality and precludes informative prediction beyond the range of the data. Finally, no comparison can be made between studies beyond overlaying plots of $k(t)$, and formal comparison with studies that use the conventional model (Eq. 3) is out of question. The last two limitations are also inherent to non-parametric nonlinear models. A model with informative parameters which can be formally compared is the way forward. Hence I here opt for the path laid out by Rovira & Rovira (2010).

Now the most important task is to find an appropriate function to describe time-variant k. Since the underlying process affecting k is detrital photosynthesis, it makes sense to start there. The change in detrital photosynthesis with time following detachment, that is detrital age, has mostly been described using various linear models (de Bettignies et al., 2020; Frontier et al., 2021; Wright and Foggo 2021; Wright et al., 2022; Wright and Kregting 2023; Wright et al., 2024). There is no reasonable intrinsic mechanism causing photosynthesis to increase following detachment. Although photosynthesis is often maintained near baseline levels it usually does not exceed this baseline. When an increase is observed it is likely due to an extrinsic factor such as a change in light environment (Frontier et al., 2021; Wright and Kregting 2023) or measurement error. Eventually photosynthesis decays to a minimum beyond which there logically is no further decline. Voilà, logistic decay! Logistic decay with detrital age was inadvertently arrived at in two independent cases: a beta regression describing Fv/Fm (Frontier et al., 2022) and a binomial regression describing the probability of photosynthesis (Wright et al., 2024). Inspired by these cases, I have formalis[illegible] logistic decay as the model of detrital photosynthesis (Wright, 2026l[illegible]

In practice, the slope and intercept of the linear term x in the standard logistic function $\frac{1}{1 + e^{-x}}$ are rarely identifiable. Non-identifiability in a linear function is typically combatted by centring the predictor, but this is not sensible when the predictor is time. Instead, the solution is to replace one of the parameters with a constant. Decay functions have a known initial value. Thus, as with exponential decay, logistic decay must have a fixed intercept. Without a fixed intercept, there is no reference point and the slope or decay rate is no longer directly comparable across cases (e.g. Wright et al., 2024). Take the concept of $t_{½}$ which is defined as the time when half of the initial value remains. Without a constant intercept, $t_{½}$ is meaningless. The problem is that unlike the exponential function, the standard logistic function is asymptotic at both bounds, so cannot take a value of 1. I believe the closest sensible value to be 0.99, which neatly translates to an intercept of 5 since $\frac{1}{1 + e^{-5}} \approx 0.99$. Because $\frac{1}{1 + e^{0}} = 0.5$ is approximately half of 0.99, the time when $x = 0$ can be interpreted as the logistic half-life, analogous to the exponential half-life. In Wright (2026b), I have therefore described detrital photosynthesis (p) as

$$p(t) = \frac{\alpha + \tau}{1 + e^{\frac{5(t - \mu)}{\mu}}} - \tau \tag{6}$$

where α and τ may be read as the maximal CO_2 assimilation and production of the plant and its detritus respectively and μ is the sigmoid inflection point, which I have termed the photosynthetic half-life (Wright, 2026b). The naming of parameters is inspired by the story of the golem (McElreath, 2020). As an entity in limbo, the golem's state is controlled by the word אמת (graecised τμα), symbolising life (א), transition (מ) and death (ת). In the form of my parameters, these letters similarly control the transition of detritus between physiological states.

In addition to modelling detrital photosynthesis, $p(t)$ can be used to describe time-variant k: $k(t) = p(t)$. I provide empirical support for this decision by comparing the performance of $p(t)$ with alternative parameterisations of the logistic function (see *Parameterisation of k(t)* in *The statistical model*). Inserting Eq. 6 in place of $k(t)$ in Eq. 4 yields

$$\frac{dm}{dt} = m\left(\frac{\alpha + \tau}{1 + e^{\frac{5(t - \mu)}{\mu}}} - \tau\right) \tag{7}$$

where α is now the initial exponential growth rate (aka relative or specific growth rate) of detritus at detachment and τ is the maximal exponential decay rate of detritus *sensu stricto* as $t \to \infty$. μ remains the photosynthetic half-life. The influence of each parameter on the trajectory of time-variant k is visualised in Fig. 1 (cf. Fig. S1).

This equation could be used directly with statistical [illegible]re that allows solving ordinary differential equations on the fly. But there is a simpler, generally applicable solution. Rovira & Rovira (2010) have shown that the logistic function can be integrated alongside the ordinary differential equation to yield a function like Eq. 3. Since we know our function is bounded by $t = 0$, we use an accumulation function. The definite integral of Eq. 6 is

$$\int_0^t p(t)dt = t\alpha + \frac{\mu(\alpha + \tau)}{5}\ln\frac{1 + e^{\frac{5(t - \mu)}{\mu}}}{1 + e^{-5}} \tag{8}$$

Thus when Eq. 7 is integrated as a whole, we get the final form

$$m(t) = m_0 e^{t\alpha + \frac{\mu(\alpha + \tau)}{5}\ln\frac{1 + e^{\frac{5(t - \mu)}{\mu}}}{1 + e^{-5}}} \tag{9}$$

Decomposition data are best analysed relative to initial mass, i.e. after normalisation. When $m_0 = 1$, the function simplifies to one with only three parameters, the now familiar α, μ and τ. The influence of each parameter on the shape of the curve m describes over time is illustrated in Fig. 1 (cf. Fig. S1). α, μ and τ may be formally compared within and between studies, including those that use the conventional model (Eq. 3). τ is the exponential decay rate of detritus *sensu stricto* once detrital photosynthesis has been factored out. Since conventional decomposition theory assumes that physiology does not play a role in detrital dynamics, k in Eq. 3 also represents the exponential decay rate of detritus *sensu stricto*. Thus τ may be directly compared to k to assess the influence of detrital photosynthesis on decomposition. Similarly, μ and $t_{½}$ have the same units and may be contrasted (Wright, 2026b). I make these formal comparisons in *Examples* below. In the next section I turn Eq. 9 into a fully functional statistical model.

3. The statistical model

3.1. Statistical framework

Maximum likelihood solutions to nonlinear modelling invariably demand the tedious task of guessing starting values. Rovira & Rovira (2010) followed this approach using the Levenberg-Marquardt algorithm which is more robust to poor starting values than the standard Gauss-Newton algorithm. Despite this advantage, they still needed to manually approximate the starting values before passing them to the algorithm. Bayesian inference using Markov chain Monte Carlo (MCMC) is more flexible in that there is no practical difference between fitting linear and parametric nonlinear models (McElreath, 2020). Rather than starting values which can make or break the model, a prior probability distribution is provided, which at worst can lead to inefficient sampling and uncertain prediction. With the increase in computing power, every laptop can now run MCMC. For these and other reasons Bayesian inference is quickly becoming the *status quo* in applied statistics across disciplines (McElreath, 2020). All models in this study are therefore Bayesian and described using standard notation *sensu* McElreath (2020).

The most popular statistical language is R (R Core Team, 2026) and the most popular Bayesian language is Stan (Carpenter et al., 2017) which integrates nicely with R through various packages. Hence I developed the statistical model within this joint framework. Specifically, I use R v4.5.2 with the tidyverse package family v2.0.0 (Wickham et al., 2019) and here v1.0.2 (Müller, 2025) within the integrated development environment RStudio v2026.01.0+392 (RStudio Team, 2026). All models in this study are written in Stan and sampled with 8 Markov chains in CmdStan v2.38.0 (Lee et al., 2026) via the R interface cmdstanr v0.9.0 (Gabry et al., 2025) with the aid of tidybayes v3.0.7 (Kay, 2024). To assess model quality, I check $\hat{R}$ convergence scores and effective

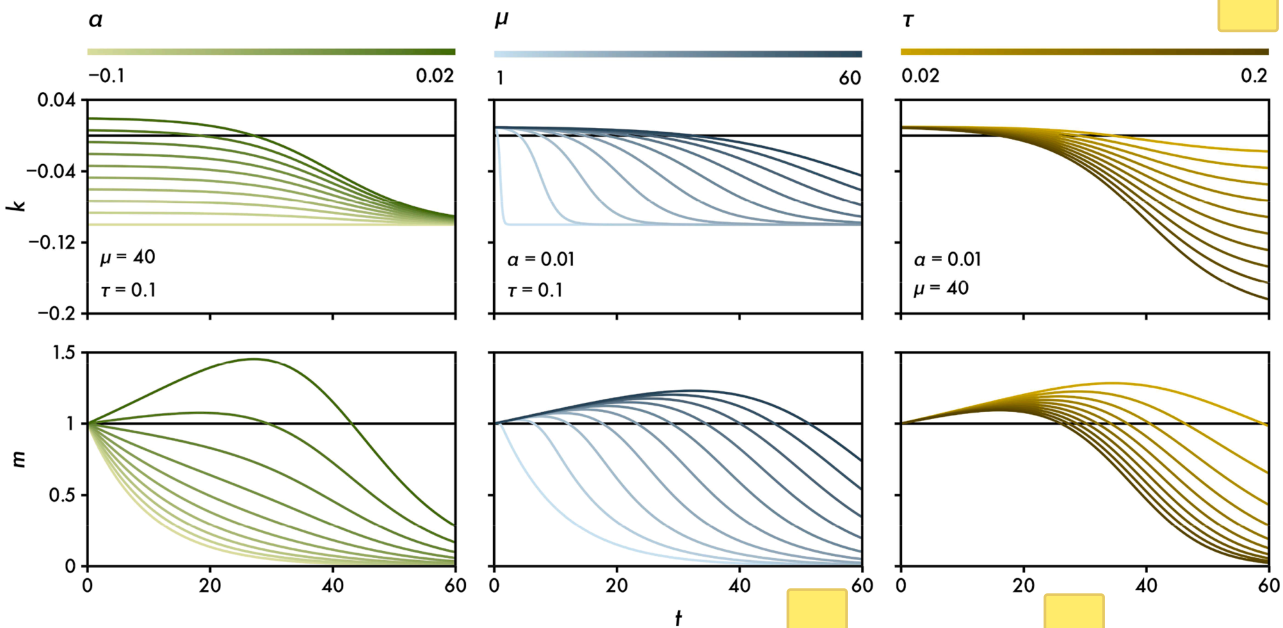

Fig. 1. The mathematical model: the effect of varying α, μ and τ on the trajectory of the exponential ~~decay~~ rate k (Eq. 6) and ~~the proportion of~~ remaining mass m (Eq. 9) over time t (cf. Fig. S1).

sample sizes and visually scrutinise trace rank and pair plots with bayesplot v1.14.0 (Gabry et al., 2019). All reported results are posterior probabilities and derived central tendencies and intervals, which are translated into figures and tables using ggh4x v0.3.1 (van den Brand, 2025), patchwork v1.3.2 (Pedersen, 2025) and officer v0.7.1 (Gohel et al., 2025). See github.com/lukaseamus/limbodeco for further details.

3.2. Parameterisation of $k(t)$

As explained above, there is a strong conceptual reason for choosing Eq. 6 to describe time-variant k. Nonetheless, I empirically tested the performance of Eq. 6 against two alternative parameterisations of the logistic function: one where the slope instead of the intercept is a constant, and another where neither intercept nor slope are constants. Unlike the intercept, the slope has no reasonable *a priori* value. For simplicity I therefore chose a value of 1 based on simulation (Fig. S1). In contrast to the constant intercept parameterisation (Eq. 6) which logically slows photosynthetic decay for large μ (Fig. 1), the constant slope parameterisation allows rapid photosynthetic decay independent of μ (Fig. S1). This is less intuitive but agrees better with observations of photosynthesis seemingly being unperturbed following detachment but decaying rapidly at some later time (de Bettignies et al., 2020; Frontier et al., 2021, 2022). The full parameterisation with no constants is equivalent to the original naïve beta and binomial models (Frontier et al., 2022; Wright et al., 2024).

Each parameterisation of $k(t)$ was tested on the two largest, most time-resolved decomposition datasets (Vandendriessche et al., 2007; Frontier et al., 2022). α, μ and τ were estimated globally for the smaller dataset ($n = 252$, Frontier et al., 2022) and stratified by replicate for the larger dataset ($n = 1651$, Vandendriessche et al., 2007), in both cases using a normal likelihood with homogenous variance and identical priors across parameterisations. I compared convergence in terms of $\widehat{R}$, effective sample size and trace rank plots, predictive performance in terms of leave-one-out cross-validation computed with Pareto-smoothed importance sampling (Vehtari et al., 2017), and identifiability with prior–posterior and pair plots. All statistics are reported in the supplement and plots are available at github.com/lukaseamus/limbodeco/tree/main/Plots. For both datasets, using Eq. 6 as $k(t)$ results in marginally better convergence (Table S1). Eq. 6 also yields the highest predictive accuracy for the small dataset, albeit the lowest for the large dataset, and the lowest level of overfitting in both cases (Table S1). As anticipated in *The mathematical model*, the logistic slope and intercept of the full parameterisation are usually non-identifiable. Without a constant, the prior therefore essentially determines either slope or intercept. For the smaller dataset, the constant slope parameterisation displays stronger posterior correlation and unstable posterior distributions. While for large datasets slope and intercept may be identifiable and the difference between parameterisations marginal, Eq. 6 is not only conceptually but also empirically the strongest parameterisation of $k(t)$ across all cases.

3.3. Likelihood function

Several hierarchical functions are needed to describe a full statistical model. The one to rule them all is the likelihood function. It describes the probability distribution of the data. Most likelihoods are described by two parameters, usually the expected value and spread. The mathematical model (Eq. 9) describes the expected value of the likelihood. But what likelihood?

Convention would favour the normal distribution but there is almost always a more fitting choice. Normalised decomposition data are ratios of final to initial mass that are positive real numbers and can exceed 1 due to growth caused by detrital photosynthesis. Several distributions are restricted to positive real numbers, such as the popular gamma and lognormal distributions. But perhaps we can do better by thinking generatively. Mass itself can only take positive real numbers and may be thought of as the sum of cells, growth increments etc. Thus the gamma distribution describes mass well. The ratio of two gamma distributions is a beta prime distribution. Technically this is only true if both gamma distributions share the same rate or scale, i.e. size of summand, and are independent. We can assume the former because both measures of mass can be broken down into the same components, e.g. cells. It is difficult to assume the latter because the fundamental assumption of the model is that the change in mass depends on mass (Eq. 1). Nonetheless, beta prime remains the best approximation of the true data distribution I can think of.

I empirically tested the validity of this choice by comparing the beta prime likelihood to the normal, lognormal and gamma likelihoods with homogenous variance and identical priors. As before, I compare convergence, predictive performance and identifiability across the two datasets. Convergence is similar across likelihoods (Table S2). Predictive performance is highest for the normal likelihood given the small dataset while the gamma and beta prime likelihoods perform best given the large dataset; the lognormal likelihood performs worst in both cases (Table S2). Since the gamma and beta prime likelihoods perform similarly and obey $m > 0$, either is a reasonable choice.

3.4. Heteroskedasticity

If Eq. 9 describes the expected value of the likelihood, what describes its spread? Under the typical assumption of homogeneity of variance, this likelihood parameter is usually estimated globally. However, variance rarely being homogenous, it is sensible to model it as a function of the predictors.

In the case of normalised decomposition, $m_0 = 1$ is predetermined, meaning there is no uncertainty at $t = 0$. But with time comes uncertainty and we expect variance to increase. Sometimes remaining mass is calculated relative to mean rather than sample-specific initial mass so there is uncertainty at $t = 0$ (e.g. Brouwer, 1996, Krumhansl and Scheibling, 2012), but even then it naturally increases with time. The beta prime distribution, typically parameterised by the two shape parameters α and β, is most intuitively parameterised in terms of mean $\mu = \frac{\alpha}{\beta - 1}$ and precision $\nu = \beta - 2$, where variance $\sigma^2 = \frac{\mu(1 + \mu)}{\nu}$ (Bourguignon et al., 2021). To reparameterise the likelihood, α and β need to be expressed in terms of μ and ν as $\alpha = \mu(1 + \nu)$ and $\beta = 2 + \nu$. Since ν is inversely related to σ^2, it is expected to decrease over time but cannot violate $\nu > 0$ so an appropriate function is

$$\nu(t) = (\epsilon - \theta)e^{-\lambda t} + \theta \tag{10}$$

where ϵ is ν at $t = 0$, λ is the rate of exponential decay in ν, and θ is ν as $t \to \infty$, inspired by its representation of Thánatos. This decay function can be used to describe precision in any likelihood while a saturating function would be the logical choice for describing variance.

The improvement of modelling heteroskedasticity over assuming homoskedasticity was empirically tested for the beta prime and gamma likelihoods. The gamma distribution may be parameterised in terms of scale or rate. The rate is inversely related to scale and variance and can therefore be described by Eq. 10. In all cases the heteroskedastic models outperform the homoscedastic ones with the gamma likelihood performing marginally better (Table S3).

3.5. Optimal model

Empirically speaking, the gamma and beta prime likelihoods are effectively interchangeable with gamma performing better in some cases. Conceptually I prefer the beta prime distribution and use it as the likelihood hereinafter. Combining the beta prime likelihood and Eqs. 9 and 10 yields the full statistical model

$$m_i \sim \text{Beta}'\left(\overline{m}_i(1 + \nu_i) , 2 + \nu_i \right) \tag{11}$$

$$\overline{m}_i = e^{t_i \alpha + \frac{\mu(\alpha + \tau)}{5} \ln \frac{1 + e^{\frac{5(t_i - \mu)}{\mu}}}{1 + e^{-5}}}$$

$$\nu_i = (\epsilon - \theta)e^{-\lambda t_i} + \theta$$

$$\alpha,\ \mu,\ \tau,\ \epsilon,\ \lambda,\ \theta \sim \text{Priors}$$

where m_i is the ratio of final to initial mass observed at t_i, $\overline{m}_i$ and ν_i being the mean and precision of m_i. $\overline{m}_i$ and ν_i are described by Eqs. 9 and 10, which for readability I will hereinafter refer to as $m(t, \alpha, \mu, \tau)$ and $\nu(t, \epsilon, \lambda, \theta)$.

Bayesian inference requires a prior probability distribution for each parameter (McElreath, 2020). All parameters except α are restricted to positive values and must therefore be assigned a prior that respects this constraint. α may have a normal prior allowing it to take positive (initial growth) or negative (initial decomposition) values but in some cases it may be desirable to enforce initial growth by assigning a prior with that constraint. Beyond simple constraints, priors must be chosen based on existing empirical and/or conceptual knowledge and simulation (McElreath, 2020). α, μ and τ are biologically meaningful and can thus be strongly informed by the literature. ϵ, λ and θ are more abstract. Based on simulation I found $\epsilon \sim \text{Gamma}(4, 10^{-4})$, $\lambda \sim \text{Exponential}(1)$ and $\theta \sim \text{Gamma}(4, 0.008)$ to be sensible.

Multilevel modelling is the state of the art because it allows estimation of hyperparameters and prediction for unobserved categories, deals with data sparsity, sampling bias and pseudo-replication, and regularises spurious effects (McElreath 2020). Eq. 11 can easily be turned into a multilevel model by partially pooling some or all parameters and can thus be used in meta-analysis. For μ, τ, ϵ, λ and θ, partial pooling is best achieved in log space because the normal distribution is easiest to sample when nested and allows non-centred parameterisation.

In the remainder of this paper, I show that for most macroalgal decomposition data Eq. 11 outperforms the conventional model

$$m_i \sim \text{Normal}\left(\overline{m}_i , \sigma \right) \tag{12}$$

$$\overline{m}_i = e^{-kt_i}$$

$$k,\ \sigma \sim \text{Priors}$$

where $\overline{m}_i$ is described by Eq. 3. A normal likelihood and homogenous variance are conventional assumptions, so they form my baseline. Given the small and large test datasets, Eq. 11 respectively has similar and better predictive accuracy than Eq. 12 (Table S4). In *Examples*, I apply Eqs. 11 and 12 to five ecological effects in seven published studies (Bourguès et al., 1996; Brouwer, 1996; Vandendriessche et al., 2007; Hamersley et al., 2015; de Bettignies et al., 2020; Frontier et al., 2021, 2022). I either received data from the lead author (Vandendriessche et al., 2007; Hamersley et al., 2015; de Bettignies et al., 2020; Frontier et al., 2021, 2022) or digitised figures using WebPlotDigitizer v4.6 (Rohatgi, 2022). Parameter estimates for both models are presented in Table 1 and contrasts between comparable parameters across models are shown in Table 2.

3.6. Sampling efficiency

There are some final considerations for Eq. 11 that are purely computational. Stan uses Hamiltonian Monte Carlo to explore probability space. Some of the expressions I have used in Eqs. 9 and 10 are easier to read but not friendly to the MCMC sampler. These can be improved for computational efficiency using the quotient and product rules of logarithms:

$$m(t) = e^{t\alpha + \frac{\mu(\alpha + \tau)}{5}\left(\ln\left(1 + e^{\frac{5(t - \mu)}{\mu}}\right) - \ln\left(1 + e^{-5}\right)\right)} \tag{13}$$

$$\nu(t) = e^{\ln(\epsilon - \theta) - \lambda t} + \theta \tag{14}$$

The $\ln(1 + e^x)$ expressions in Eq. 12 are further simplified in Stan using the wrapper function log1p_exp which is numerically more stable. An equivalent function does not exist in R but can be written as `log1p_exp <- function(x) ifelse( x > 0, x + log1p( exp( - x) ), log1p( exp(x) ) )`. Note that Eq. 10 performs better than Eq. 14 in some cases, e.g. when prior variance for ϵ and/or θ is large, which results in $\epsilon \leq \theta$ at initialisation, Eq. 14 causes the log probability to evaluate to $-\infty$.

4. Examples

4.1. Dead or alive

Brouwer (1996) provides a nice example of detrital photosynthesis

Table 1
Macroalgal decomposition parameter estimates. Parameters of the presented macroalgal model (α, μ, τ in Eq. 11) are inferred from seven empirical datasets alongside the conventional decay constant k (Eq. 12). Estimates are mean ± standard deviation of 8×10^4 posterior samples; the median is given in parentheses when k is highly right-skewed. NB exponential rates are given as percentages to improve readability.

	k (% day^{-1})	α (% day^{-1})	μ (days)	τ (% day^{-1})
Brouwer (1996)				
Desmarestia anceps				
Control	0.06 ± 0.014	0.068 ± 0.032	585 ± 100	5.6 ± 1.5
Pre-killed	3.4 ± 0.29	−1.3 ± 0.5	30 ± 12	5.6 ± 1.5
Hamersley et al. (2015)				
Macrocystis pyrifera				
Fresh	5.7 ± 0.43	−0.3 ± 0.64	7 ± 1.1	11 ± 1.1
Senescent	10 ± 0.9	−0.79 ± 1.6	1.6 ± 0.56	12 ± 0.64
Detached	7.1 ± 0.57	−0.99 ± 1.6	2.9 ± 1.8	9.1 ± 0.8
de Bettignies et al. (2020)				
Laminaria hyperborea				
Fresh (2017)	1.1 ± 0.1	0.26 ± 0.38	16 ± 9.5	1.3 ± 0.1
Senescent (2017)	3.7 ± 2.2	−2.4 ± 0.57	34 ± 33	3.6 ± 0.54
Senescent (2018)	0.54 ± 0.06	0.15 ± 0.34	71 ± 23	1.6 ± 0.39
Bourguès et al. (1996)				
Ulvaria obscura				
Spring	3.8 ± 1.2	0.5 ± 0.14	13 ± 3.2	14 ± 6.9
Summer	7.7 ± 3.9	0.5 ± 0.14	4 ± 2.9	12 ± 7.5
Autumn	2.5 ± 0.87	0.51 ± 0.14	23 ± 3.4	14 ± 4.8
Winter	2.4 ± 0.97	0.49 ± 0.13	24 ± 19	11 ± 7.9
Frontier et al. (2021)				
Laminaria hyperborea				
0 m	0.42 ± 5.8 (0.075)	0.4 ± 0.1	181 ± 190	14 ± 4.6
15 m	2.4 ± 43 (0.56)	0.4 ± 0.1	56 ± 58	14 ± 4.6
30 m	18 ± 136 (4.3)	0.4 ± 0.1	18 ± 22	14 ± 4.6
Laminaria ochroleuca				
0 m	1.5 ± 17 (0.34)	0.42 ± 0.11	169 ± 174	14 ± 4.6
15 m	2.2 ± 30 (0.57)	0.42 ± 0.11	87 ± 88	14 ± 4.6
30 m	4.7 ± 71 (0.98)	0.42 ± 0.11	46 ± 50	14 ± 4.6
Frontier et al. (2022)				
Laminaria hyperborea				
0.5 m	0.15 ± 1.7 (0.024)	0.3 ± 0.12	114 ± 20	16 ± 15
1.5 m	0.2 ± 2.5 (0.039)	0.3 ± 0.12	97 ± 16	16 ± 15
3 m	0.4 ± 3.4 (0.081)	0.3 ± 0.12	75 ± 14	16 ± 15
Laminaria ochroleuca				
0.5 m	1.1 ± 25 (0.16)	0.64 ± 0.24	75 ± 11	24 ± 23
1.5 m	1.3 ± 11 (0.25)	0.64 ± 0.24	56 ± 8	24 ± 23
3 m	2.4 ± 16 (0.5)	0.64 ± 0.24	37 ± 6.8	24 ± 23
Vandendriessche et al. (2007)				
Ascophyllum nodosum				
5°C Control	0.0074 ± 0.0045	0.42 ± 0.033	262 ± 30	4.2 ± 0.77
5°C Grazed	0.049 ± 0.026	0.42 ± 0.033	248 ± 29	8.3 ± 1.4
10°C Control	1 ± 0.48	0.42 ± 0.033	18 ± 1.9	2.8 ± 0.29
10°C Grazed	3.6 ± 1.6	0.42 ± 0.033	11 ± 1	10 ± 1

Table 1 (*continued*)

	k (% day^{-1})	α (% day^{-1})	μ (days)	τ (% day^{-1})
Brouwer (1996)				
15°C Control	1.5 ± 0.73	0.42 ± 0.033	12 ± 1.3	3.7 ± 0.42
15°C Grazed	6.5 ± 3	0.42 ± 0.033	3.5 ± 0.4	7.8 ± 0.76
18°C Control	1.6 ± 0.74	0.42 ± 0.033	16 ± 1.6	5.7 ± 0.63
18°C Grazed	7 ± 3.1	0.42 ± 0.033	3.5 ± 0.38	7.7 ± 0.73
Fucus vesiculosus				
5°C Control	0.0044 ± 0.0027	0.36 ± 0.03	751 ± 96	11 ± 2.1
5°C Grazed	0.029 ± 0.015	0.36 ± 0.03	711 ± 88	22 ± 3.7
10°C Control	0.62 ± 0.29	0.36 ± 0.03	51 ± 5.4	7.2 ± 0.95
10°C Grazed	2.1 ± 0.97	0.36 ± 0.03	31 ± 3.2	26 ± 3.5
15°C Control	0.91 ± 0.45	0.36 ± 0.03	35 ± 3.9	9.7 ± 1.4
15°C Grazed	3.9 ± 1.8	0.36 ± 0.03	9.9 ± 1.2	20 ± 2.5
18°C Control	0.95 ± 0.45	0.36 ± 0.03	46 ± 4.9	15 ± 2.2
18°C Grazed	4.1 ± 1.8	0.36 ± 0.03	9.9 ± 1.1	20 ± 2.3

influencing decomposition of the feathery brown seaweed *Desmarestia anceps* at Signy Island in Antarctica. She pre-killed some of the detritus by heating it to 50°C for 10–15 min prior to *in situ* decomposition. The resulting effect is remarkable but Brouwer (1996) was unable to quantify it due to lack of a suitable model. She only reports analysis of variance F statistics for change over time and describes the divergence in decomposition trajectories. Remaining mass at each timepoint is summarised as mean ± standard deviation of five samples. To correctly account for summarised uncertainty, I use a measurement error model which gives less predictive weight to means with larger standard deviations (McElreath, 2020). Since I assume that these data obey a beta prime distribution, it follows that measurement error is also beta prime distributed. Directly parameterising beta prime error in terms of sample mean μ and sample standard deviation σ is hard on the MCMC sampler. I therefore pre-calculate the sample precision $\nu = \frac{\mu(1+\mu)}{\sigma^2}$. The resulting model is

$$m_{\text{obs},i} \sim \text{Beta}'\left(m_i(1 + \nu_{\text{obs},i}) \, , \, 2 + \nu_{\text{obs},i} \right) \tag{15}$$

$$m_i \sim \text{Beta}'\left(\overline{m}_i(1 + \nu_i) \, , \, 2 + \nu_i \right)$$

$$\overline{m}_i = m\left(t_i \, , \, \alpha_{E_i} \, , \, \mu_{E_i} \, , \, \tau \right)$$

$$\nu_i = \nu\left(t_i \, , \, \epsilon \, , \, \lambda_{E_i} \, , \, \theta \right)$$

$$\alpha_E \sim \text{Normal}\left(-0.01 \, , \, 0.005 \right)$$

$$\mu_E \sim \text{Gamma}\left(\frac{150^2}{100^2} \, , \, \frac{150}{100^2} \right)$$

$$\tau \sim \text{Gamma}\left(\frac{0.1^2}{0.05^2} \, , \, \frac{0.1}{0.05^2} \right)$$

$$\epsilon \sim \text{Gamma}\left(4 \, , \, 10^{-4} \right)$$

$$\lambda_E \sim \text{Exponential}\left(1 \right)$$

$$\theta \sim \text{Gamma}\left(4 \, , \, 0.008 \right)$$

where m_i is the latent variable of unobserved true values, subscript obs indicates observed variables and E is the experimental treatment variable. τ and θ are not stratified by E since the experimental duration is not long enough to identify these parameters for the control treatment and it is reasonable to assume that decomposition rates converge once detrital photosynthesis has ceased. Predictions by this model are visualised in

Table 2
Comparison between macroalgal (Eq. 11) and conventional (Eq. 12) decomposition parameters. Log ratios indicate by how many orders of magnitude the dead decay rate τ is greater than the conventional decay rate k, or the detrital half-life $t_{½}$ is greater than the photosynthetic half-life μ. P gives the posterior probability of these differences. Prior $P(\tau > k) = 0.5$, equivalent to a coin toss, but prior $P(t_{½} > \mu)$ could not be set in a similar manner because $t_{½}$ is derived from k. Estimates are mean ± standard deviation of 8×10^4 posterior samples.

	$\log_{10}(\tau / k)$	$P(\tau > k)$	$\log_{10}(t_{½} / \mu)$	$P(t_{½} > \mu)$
Brouwer (1996)				
Desmarestia anceps				
Control	2 ± 0.14	1	0.32 ± 0.12	0.99
Pre-killed	0.2 ± 0.11	0.98	−0.14 ± 0.16	0.17
Hamersley et al. (2015)				
Macrocystis pyrifera				
Fresh	0.3 ± 0.052	1	0.25 ± 0.08	1
Senescent	0.068 ± 0.045	0.93	0.65 ± 0.13	1
Detached	0.11 ± 0.05	0.99	0.58 ± 0.2	0.99
de Bettignies et al. (2020)				
Laminaria hyperborea				
Fresh (2017)	0.082 ± 0.053	0.94	0.64 ± 0.15	1
Senescent (2017)	0.027 ± 0.17	0.6	−0.086 ± 0.37	0.42
Senescent (2018)	0.47 ± 0.11	1	0.29 ± 0.17	0.99
Bourguès et al. (1996)				
Ulvaria obscura				
Spring	0.55 ± 0.2	1	0.19 ± 0.18	0.88
Summer	0.2 ± 0.19	0.88	0.47 ± 0.31	0.95
Autumn	0.75 ± 0.19	1	0.12 ± 0.16	0.78
Winter	0.61 ± 0.33	0.95	0.2 ± 0.34	0.73
Frontier et al. (2021)				
Laminaria hyperborea				
0 m	2.3 ± 0.78	1	0.87 ± 0.85	0.85
15 m	1.4 ± 0.7	0.98	0.5 ± 0.78	0.75
30 m	0.5 ± 0.73	0.76	0.13 ± 0.82	0.56
Laminaria ochroleuca				
0 m	1.6 ± 0.72	0.98	0.24 ± 0.79	0.62
15 m	1.4 ± 0.69	0.98	0.3 ± 0.77	0.65
30 m	1.1 ± 0.73	0.94	0.36 ± 0.8	0.68
Frontier et al. (2022)				
Laminaria hyperborea				
0.5 m	2.7 ± 0.87	1	1.4 ± 0.81	0.97
1.5 m	2.5 ± 0.86	1	1.3 ± 0.79	0.96
3 m	2.2 ± 0.88	1	1.1 ± 0.81	0.92
Laminaria ochroleuca				
0.5 m	2 ± 0.84	0.99	0.77 ± 0.77	0.85
1.5 m	1.8 ± 0.83	0.99	0.7 ± 0.75	0.84
3 m	1.6 ± 0.85	0.97	0.59 ± 0.77	0.79
Vandendriessche *et al.* (2007)				
Ascophyllum nodosum				
5°C Control	2.8 ± 0.26	1	1.6 ± 0.25	1
5°C Grazed	2.3 ± 0.23	1	0.81 ± 0.23	1
10°C Control	0.46 ± 0.2	0.99	0.61 ± 0.2	1
10°C Grazed	0.49 ± 0.19	0.99	0.3 ± 0.19	0.94
15°C Control	0.43 ± 0.2	0.98	0.62 ± 0.2	1
15°C Grazed	0.11 ± 0.19	0.72	0.53 ± 0.2	1
18°C Control	0.59 ± 0.2	1	0.47 ± 0.2	0.99
18°C Grazed	0.081 ± 0.19	0.66	0.5 ± 0.19	1
Fucus vesiculosus				
5°C Control	3.5 ± 0.27	1	1.4 ± 0.26	1
5°C Grazed	2.9 ± 0.24	1	0.59 ± 0.23	1
10°C Control	1.1 ± 0.2	1	0.39 ± 0.2	0.97
10°C Grazed	1.1 ± 0.2	1	0.079 ± 0.2	0.65
15°C Control	1.1 ± 0.21	1	0.4 ± 0.21	0.97
15°C Grazed	0.76 ± 0.2	1	0.31 ± 0.2	0.94
18°C Control	1.2 ± 0.21	1	0.24 ± 0.2	0.88
18°C Grazed	0.73 ± 0.19	1	0.28 ± 0.19	0.93

Fig. 2 (cf. Fig. S2). Predictions for the mean and new observations of m are respectively denoted by $\overline{m}$ and $\tilde{m}$. Summaries of parameter distributions are given in Table 1.

The model suggests that pre-killed *D. anceps* detritus decomposes much faster than the control (Fig. 2). Untreated detritus has an initial relative growth rate of 0.068 ± 0.032 % day^{-1} (mean ± standard deviation, $P_{\alpha>0} = 0.99$) while pre-killed detritus has an initial decay rate of 1.3 ± 0.5 % day^{-1} ($P_{\alpha<0} = 0.99$) (Table 1). The photosynthetic half-life of *D. anceps* detritus is 585 ± 100 days and without detrital photosynthesis it decomposes at a rate of 5.6 ± 1.5 % day^{-1} (Table 1). The conventional model (Fig. S2) underestimates decomposition by two orders of magnitude; pre-killing detritus ameliorates this difference, but only partly (Tables 1 and 2). These results suggest that detrital photosynthesis confounds the conventional model and even after experimental removal of physiology, it may not yield reliable predictions for detritus *sensu stricto*.

4.2. Senescence

Hamersley et al. (2015) and de Bettignies et al. (2020) both studied the effect of tissue senescence on kelp decomposition. Hamersley et al. (2015) collected attached fresh and senescent fronds of the giant kelp *Macrocystis pyrifera* at Catalina Island in California along with naturally detached wrack and raft material which they regarded as detritus *sensu stricto*. de Bettignies et al. (2020) collected old and new blades from above and below the abscission zone of the boreal kelp *Laminaria hyperborea* at Roscoff in Brittany. Their old blade thus represents attached senescing tissue that the plant is in the process of abscising. Hamersley et al. (2015) refrained from fitting a model but de Bettignies et al. (2020) fit Eq. 3 using maximum likelihood estimation with the Gauss-Newton algorithm, which passably describes their published data from 2017, and report k = 3.66 and 1.07 % day^{-1} for old and new blades. Notably, they excluded data from a second experiment on old blades at the same site the following year (2018) which cannot be described by Eq. 3. With the permission of the lead author, I include these data here to show that they present no problem for the proposed macroalgal decomposition model. The model for Hamersley et al. (2015) is

$$m_i \sim \mathrm{Beta}'(\, \overline{m}_i(1 + \nu_i) \,,\, 2 + \nu_i \,) \tag{16}$$

$$\overline{m}_i = m(\, t_i \,,\, \alpha_{A_i} \,,\, \mu_{A_i} \,,\, \tau_{A_i} \,)$$

$$\nu_i = \nu(\, t_i \,,\, \epsilon \,,\, \lambda \,,\, \theta \,)$$

$$\alpha_A \sim \mathrm{Normal}(\, \overline{\alpha} \,,\, \sigma_\alpha \,)$$

$$\ln\mu_A \sim \mathrm{Normal}(\, \overline{\ln\mu} \,,\, \sigma_{\ln\mu} \,)$$

$$\ln\tau_A \sim \mathrm{Normal}(\, \overline{\ln\tau} \,,\, \sigma_{\ln\tau} \,)$$

$$\overline{\alpha} \sim \mathrm{Normal}(\, 0 \,,\, 0.01 \,)$$

$$\overline{\ln\mu} \sim \mathrm{Normal}(\, \ln 10 \,,\, 0.5 \,)$$

$$\overline{\ln\tau} \sim \mathrm{Normal}(\, \ln 0.2 \,,\, 0.5 \,)$$

$$\sigma_\alpha \sim \mathrm{Normal}_{0,\infty}(\, 0 \,,\, 0.01 \,)$$

$$\sigma_{\ln\mu},\ \sigma_{\ln\tau} \sim \mathrm{Normal}_{0,\infty}(\, 0 \,,\, 0.5 \,)$$

$$\epsilon,\ \lambda,\ \theta \sim \text{Priors as in Eq. 14}$$

where all parameters describing $\overline{m}_i$ are partially pooled across tissue age A. Predictions are shown in Fig. 3a (cf. Fig. S3a). The model for de Bettignies et al. (2020) is

$$m_i \sim \mathrm{Beta}'(\, \overline{m}_i(1 + \nu_i) \,,\, 2 + \nu_i \,) \tag{17}$$

$$\overline{m}_i = m(\, t_i \,,\, \alpha_{A_i} \,,\, \mu_{A_i} \,,\, \tau_{A_i} \,)$$

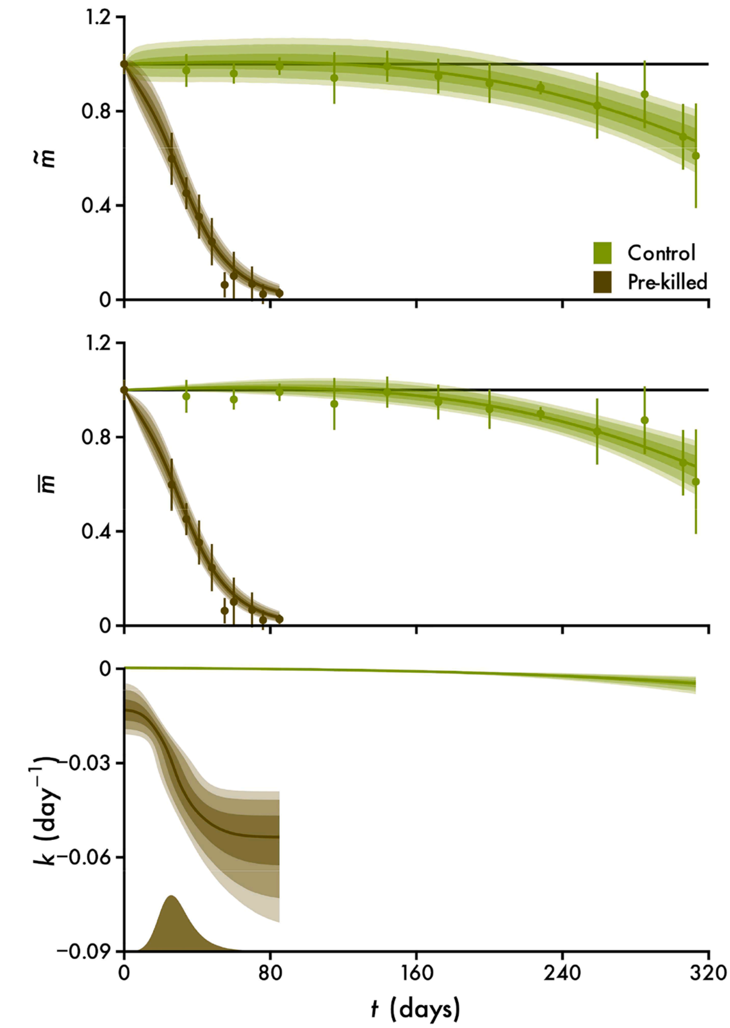

Fig. 2. Example 1: the effect of physiology on the decomposition of *Desmarestia anceps*, shown as normalised remaining mass (mean ± standard deviation, $n = 5$) over time t ($n = 115$, data from Brouwer 1996). Data are plotted alongside predictions of new observations $\tilde{m}$, the mean $\overline{m}$, the time-variant exponential ~~decay~~ rate k and the photosynthetic half-life μ (Eq. 15, cf. Fig. S2). Lines and intervals are medians and the central 50, 80 and 90% of posterior probability. The distribution is a kernel density estimate of the probability density function of μ.

$$\nu_i = \nu(\, t_i \,,\, \epsilon \,,\, \lambda \,,\, \theta \,)$$

$$\alpha_{A_i} = \delta_{A_i} - \tau_{A_i}$$

$$\ln\delta_A \sim \text{Normal}(\, \overline{\ln\delta} \,,\, \sigma_{\ln\delta} \,)$$

$$\ln\mu_A \sim \text{Normal}(\, \overline{\ln\mu} \,,\, \sigma_{\ln\mu} \,)$$

$$\ln\tau_A \sim \text{Normal}(\, \overline{\ln\tau} \,,\, \sigma_{\ln\tau} \,)$$

$$\overline{\ln\delta} \sim \text{Normal}(\, \ln 0.01 \,,\, 0.2 \,)$$

$$\overline{\ln\mu} \sim \text{Normal}(\, \ln 30 \,,\, 0.3 \,)$$

$$\overline{\ln\tau} \sim \text{Normal}(\, \ln 0.02365 \,,\, 0.5 \,)$$

$$\sigma_{\ln\delta} \sim \text{Normal}_{0,\infty}(\, 0 \,,\, 0.2 \,)$$

$$\sigma_{\ln\mu} \sim \text{Normal}_{0,\infty}(\, 0 \,,\, 0.3 \,)$$

$$\sigma_{\ln\tau} \sim \text{Normal}_{0,\infty}(\, 0 \,,\, 0.5 \,)$$

$$\epsilon,\ \lambda,\ \theta \sim \text{Priors as in Eq. 14}$$

where parameters are partially pooled as in Eq. 16, but with the added constraint $-\alpha < \tau$ since decomposition rate cannot conceivably decrease according to the theory of detrital photosynthesis. The prior for τ is centred on the mean of reported k. Predictions are shown in Fig. 3b (cf. Fig. S3b).

Both models suggest that senescent fronds tend to decompose faster than fresh ones (Fig. 3), but this effect is variable and not as strong as for pre-killed tissue (cf. Fig. 2). Detritus of *M. pyrifera* does not grow regardless of tissue age ($P_{\alpha<0} = 0.7$–0.74). New *L. hyperborea* blades grow at detachment ($P_{\alpha>0} = 0.79$) but old blades either grow ($P_{\alpha>0} = 0.67$) or decompose ($P_{\alpha<0} = 1$) (Table 1). Photosynthetic half-life of fresh *M. pyrifera* is 4.1 ± 2 days ($P_{\Delta\mu>0} = 0.95$) and 5.4 ± 1.2 days ($P_{\Delta\mu>0} = 1$) longer than for detached and senescent fronds (Fig. 3a, Table 1). But detrital photosynthesis of fresh *L. hyperborea* may cease earlier ($\Delta\mu = 55 \pm 25$ days, $P_{\Delta\mu>0} = 0.99$) than for senescent tissue (Fig. 3b, Table 1). These results suggest that senescent kelp typically has impaired photosynthesis and decomposes faster as a consequence. However, the viability of apparently senescent tissue may be highly unpredictable, as is evidenced by the change in decomposition of old *L. hyperborea* blades from one year to the next. The conventional model (Fig. S3) again underestimates decomposition in all cases and conventional estimates are most accurate when detrital photosynthesis is weakest, as for senescent *L. hyperborea* in the 2017 experiment (Tables 1 and 2).

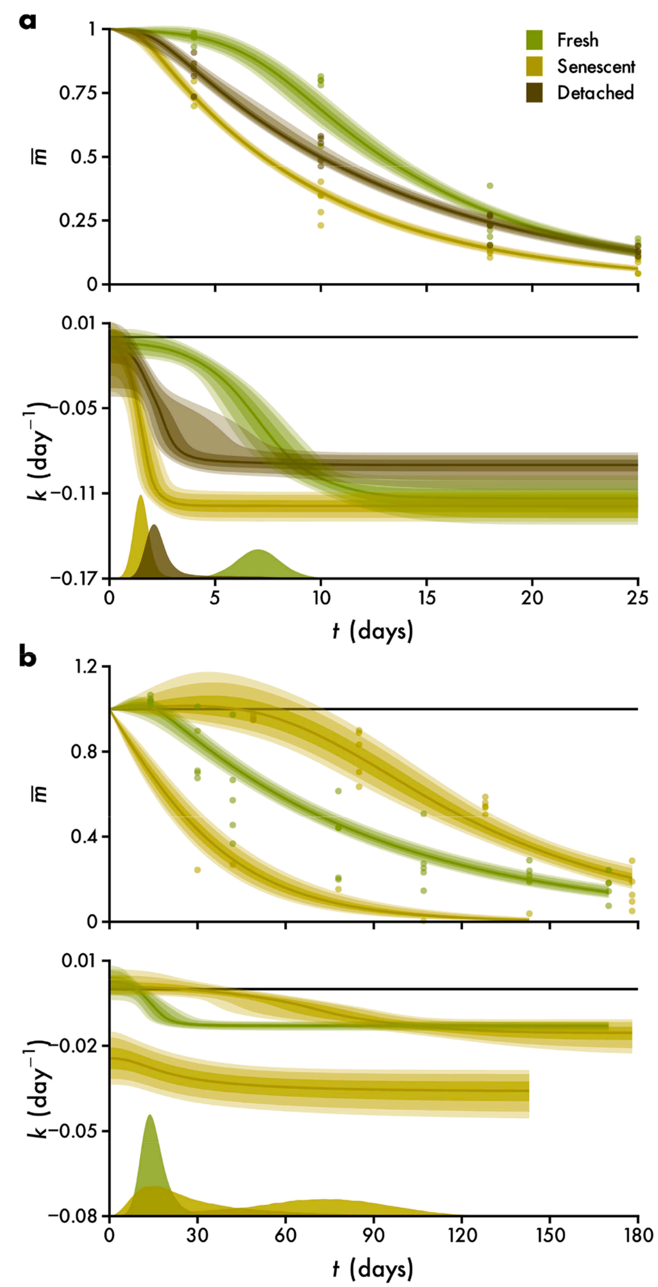

Fig. 3. Example 2: the effect of senescence on the decomposition of *Macrocystis pyrifera* (a, $n = 61$, data from Hamersley et al. 2015) and *Laminaria hyperborea* (b, $n = 63$, data from de Bettignies et al. 2020), shown as normalised remaining mass over time t. Data are plotted alongside predictions of the mean $\overline{m}$, the time-variant exponential ~~decay~~ rate k and photosynthetic half-life μ (Eqs. 16 and 17, cf. Fig. S3). [illegible]nd intervals are medians and the central 50, 80 and 90% of posterior probability. Distributions are kernel density estimates of the probability density function of μ.

4.3. Season

Bourguès et al. (1996) provide perhaps the only data of macroalgal decomposition across all seasons. They studied decomposition of the ephemeral green seaweed *Ulvaria obscura* in the intertidal Arcachon Bay in France. Bourguès et al. (1996) collected fresh *U. obscura* but buried it below 1 cm of sediment where anoxia apparently persisted and there clearly was no light for detrital photosynthesis. Given these experimental conditions, it would be surprising to see decomposition affected by physiology. Yet Bourguès et al. (1996) were faced with the challenge of having to model decomposition trajectories of sigmoidal appearance for which they chose the logistic function. Excised macroalgal tissue can survive exceptional periods of darkness (Wright et al., 2024) so it is possible that saprotrophs were initially inhibited by chemical defence despite the absence of photosynthesis. Even if this was not the case because anoxia immediately killed the seaweed, and the model therefore is conceptually wrong, it is still useful because it avoids having to fit fundamentally incomparable models as Bourguès et al. (1996) have done. Thus the model is

$$m_i \sim \mathrm{Beta}'(\, \overline{m}_i(1 + \nu_i)\,,\, 2 + \nu_i\,) \tag{18}$$

$$\overline{m}_i = m(\, t_i\,,\, \alpha_{M_i}\,,\, \mu_{M_i}\,,\, \tau_{M_i}\,)$$

$$\nu_i = \nu(\, t_i\,,\, \epsilon\,,\, \lambda_{M_i}\,,\, \theta_{M_i}\,)$$

$$\ln\alpha_M \sim \mathrm{Normal}(\, \overline{\ln\alpha}\,,\, \sigma_{\ln\alpha}\,)$$

$$\ln\mu_M \sim \mathrm{Normal}(\, \overline{\ln\mu}\,,\, \sigma_{\ln\mu}\,)$$

$$\ln\tau_M \sim \mathrm{Normal}(\, \overline{\ln\tau}\,,\, \sigma_{\ln\tau}\,)$$

$$\ln\lambda_M \sim \mathrm{Normal}(\, \overline{\ln\lambda}\,,\, \sigma_{\ln\lambda}\,)$$

$$\ln\theta_M \sim \mathrm{Normal}(\, \overline{\ln\theta}\,,\, \sigma_{\ln\theta}\,)$$

$$\overline{\ln\alpha} \sim \mathrm{Normal}(\, \ln 0.005\,,\, 0.2\,)$$

$$\overline{\ln\mu} \sim \mathrm{Normal}(\, \ln 15\,,\, 0.5\,)$$

$$\overline{\ln\tau} \sim \mathrm{Normal}(\, \ln 0.1\,,\, 0.5\,)$$

$$\epsilon \sim \mathrm{Gamma}(\, 4\,,\, 10^{-4}\,)$$

$$\overline{\ln\lambda} \sim \mathrm{Normal}(\, \ln 1\,,\, 0.4\,)$$

$$\overline{\ln\theta} \sim \mathrm{Normal}(\, \ln 500\,,\, 0.4\,)$$

$$\sigma_{\ln\alpha} \sim \mathrm{Normal}_{0,\infty}(\, 0\,,\, 0.2\,)$$

$$\sigma_{\ln\mu},\ \sigma_{\ln\tau} \sim \mathrm{Normal}_{0,\infty}(\, 0\,,\, 0.5\,)$$

$$\sigma_{\ln\lambda},\ \sigma_{\ln\theta} \sim \mathrm{Normal}_{0,\infty}(\, 0\,,\, 0.4\,)$$

where parameters describing $\overline{m}_i$ and ν_i are partially pooled across months M with the added constraint $\alpha > 0$ since, unlike the previous cases, Bourguès et al. (1996) used only fresh tissue. Predictions are shown in Fig. 4 (cf. Fig. S4).

According to the model decomposition is generally faster in warmer seasons (Fig. 4). Initial growth rate is not different between any two seasons ($P_{\Delta\alpha>0} = 0.51$–$0.56$, 0.5 being a coin toss), but in summer photosynthetic half-life is 8.8 ± 3.9 days ($P_{\Delta\mu>0} = 0.99$), 19 ± 4.2 days ($P_{\Delta\mu>0} = 1$) and 20 ± 19 days ($P_{\Delta\mu>0} = 0.97$) shorter than in spring, autumn and winter respectively (Table 1). Unless we believe the experimental conditions immediately killed the seaweed, this agrees well with temperature sensitivity of detrital photosynthesis (Wright et al., 2024) resulting in faster decomposition at higher temperatures (Filbee-Dexter et al., 2022; Frontier et al., 2022; Wright et al., 2024). As with the previous cases, the conventional model (Fig. S4) underestimates decomposition (Tables 1 and 2).

4.4. Light

The accelerating effect of depth on decomposition has repeatedly been evidenced for *Laminaria* species (Frontier et al., 2021, 2022). Physiological measurements have clarified that light limitation

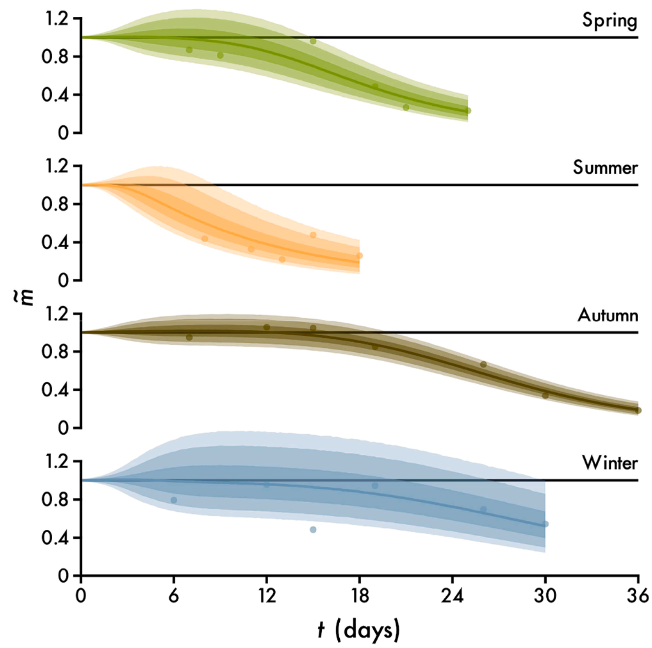

Fig. 4. Example 3: the effect of season on the decomposition of *Ulvaria obscura*, shown as normalised remaining mass over time t (n = 28, data from Bourguès et al. 1996). Data are plotted alongside predictions of new observations $\widetilde{m}$ (Eq. 18, cf. Fig. S4). Lines and intervals are medians and the central 50, 80 and 90% of posterior probability.

underlies this effect (Frontier et al., 2021, 2022; Wright et al., 2024) and detrital photosynthesis has therefore been suggested as a general mechanism opposing photodegradation (Wright 2026b). Detritus of cold temperate *L. hyperborea* and its warm temperate congener *L. ochroleuca* was either exposed to three simulated depths *ex situ* at Roscoff (Frontier et al., 2021) or deployed at three depths *in situ* in Queen Anne's Battery marina in Plymouth (Frontier et al., 2022). In both studies, samples were repeatedly measured over time. In the original permutational analysis of variance or linear multilevel model, pseudo-replication was counteracted by partially pooling replicates (Frontier et al., 2021, 2022). In a Bayesian framework, this strategy can seamlessly be applied to the nonlinear model

$$m_i \sim \mathrm{Beta}'(\ \overline{m}_i(1 + \nu_i)\ ,\ 2 + \nu_i\ ) \tag{19}$$

$$\overline{m}_i = m(\ t_i\ ,\ \alpha_i\ ,\ \mu_i\ ,\ \tau_i\ )$$

$$\nu_i = \nu(\ t_i\ ,\ \epsilon\ ,\ \lambda\ ,\ \theta\ )$$

$$\ln\alpha_i = \ln\alpha_{S_i} + \ln\alpha_{R_i}$$

$$\ln\mu_i = \ln\mu_{S_i} + \ln\mu_{R_i} - \beta_{S_i}D_i$$

$$\ln\tau_i = \ln\tau_{S_i} + \ln\tau_{R_i}$$

$$\ln\alpha_S \sim \mathrm{Normal}(\ \overline{\ln\alpha}\ ,\ \sigma_{\ln\alpha_S}\ )$$

$$\ln\alpha_R \sim \mathrm{Normal}(\ 0\ ,\ \sigma_{\ln\alpha_R}\ )$$

$$\ln\mu_S \sim \mathrm{Normal}(\ \overline{\ln\mu}\ ,\ \sigma_{\ln\mu_S}\ )$$

$$\ln\mu_R \sim \mathrm{Normal}(\ 0\ ,\ \sigma_{\ln\mu_R}\ )$$

$$\ln\beta_S \sim \mathrm{Normal}(\ \overline{\ln\beta}\ ,\ \sigma_{\ln\beta}\ )$$

$$\ln\tau_S \sim \mathrm{Normal}(\ \overline{\ln\tau}\ ,\ \sigma_{\ln\tau_S}\ )$$

$$\ln\tau_R \sim \mathrm{Normal}(\ 0\ ,\ \sigma_{\ln\tau_R}\ )$$

$$\overline{\ln\alpha} \sim \mathrm{Normal}(\ \ln 0.004\ ,\ 0.2\ )$$

$$\overline{\ln\mu} \sim \mathrm{Normal}(\ \ln 100\ ,\ 0.15\ )\ \text{or}\ \mathrm{Normal}(\ \ln 100\ ,\ 0.05\ )$$

$$\overline{\ln\beta} \sim \mathrm{Normal}(\ \ln 0.05\ ,\ 0.25\ )\ \text{or}\ \mathrm{Normal}(\ \ln 0.25\ ,\ 0.2\ )$$

$$\overline{\ln\tau} \sim \mathrm{Normal}(\ \ln 0.12\ ,\ 0.2\ )$$

$$\sigma_{\ln\alpha_S},\ \sigma_{\ln\alpha_R},\ \sigma_{\ln\tau_S},\ \sigma_{\ln\tau_R} \sim \mathrm{Normal}_{0,\infty}(\ 0\ ,\ 0.2\ )$$

$$\sigma_{\ln\mu_S},\ \sigma_{\ln\mu_R} \sim \mathrm{Normal}_{0,\infty}(\ 0\ ,\ 0.15\ )\ \text{or}\ \mathrm{Normal}_{0,\infty}(\ 0\ ,\ 0.05\ )$$

$$\sigma_{\ln\beta} \sim \mathrm{Normal}_{0,\infty}(\ 0\ ,\ 0.25\ )\ \text{or}\ \mathrm{Normal}_{0,\infty}(\ 0\ ,\ 0.2\ )$$

$$\epsilon,\ \lambda,\ \theta \sim \text{Priors as in Eq. 14}$$

where parameters describing $\overline{m}_i$ are partially pooled across species S and replicates R. Additionally, the natural logarithm of μ is modelled to decrease linearly with depth D at a species-specific rate β. D is only included in the equation of μ since it logically cannot affect initial growth or final decay. Again $\alpha > 0$ since fresh tissue was used. I applied Eq. 19 to both studies with mostly identical priors; where priors differ, they are listed first for Frontier et al. (2021) and then for Frontier et al. (2022). Mean predictions for each replicate and across replicates are shown in Fig. 5 (cf. Fig. S5).

Both models quantify how depth accelerates decomposition by reducing photosynthetic half-life. In the first study, the exponential decay rate of photosynthetic half-life with depth (β) is 7.8 ± 1.5 % m^{-1} for *L. hyperborea* and 4.5 ± 1.1 % m^{-1} for *L.ochroleuca*. In the second study, β is greater for both species—17 ± 5.5 % m^{-1} for *L.hyperborea* and 28 ± 6.4 % m^{-1} for *L.ochroleuca*—likely because light attenuation in the turbid marina (Frontier et al., 2022) is much faster than that simulated by the light filters (Frontier et al., 2021). A species reversal in sensitivity to light limitation is evident from one study to the next. Photosynthetic half-life decays 3.3 ± 1.9 % m^{-1} faster for *L.hyperborea* ($P_{\Delta\beta>0} = 0.97$) in Roscoff (Fig. 5a), while it decays 11 ± 9.5 % m^{-1} faster for *L.ochroleuca* ($P_{\Delta\beta>0} = 0.9$) in Plymouth (Fig. 5b). This reversal may be an effect of latitude, giving each species a thermal "home advantage". The interspecific contrast seems to be mostly due to detrital photosynthesis, with similar ($\Delta\tau = 0.18 \pm 6.3$ % day^{-1}, $P_{\Delta\tau>0} = 0.52$) or marginally faster ($\Delta\tau = 8.2 \pm 28$ % day^{-1}, $P_{\Delta\tau>0} = 0.63$) decay of *L.ochroleuca* detritus *sensu stricto* (Table 1). This aligns with other observations on faster decomposition of *L.ochroleuca* (Wright et al., 2022). The conventional model (Fig. S5) underestimates shallow decomposition more than deep decomposition (Tables 1 and 2), as is expected due to the declining influence of detrital photosynthesis with depth.

4.5. Temperature

Several seaweeds float after detachment and therefore decompose in a light environment that is conducive to detrital photosynthesis. Examples include seaweeds with pneumatocysts such as the giant kelp *M. pyrifera* (Rothäusler et al., 2018) and those with a honeycomb thallus such as the bull kelp *Durvillaea antarctica* (Tala et al., 2019). Vandendriessche et al. (2007) report the most resolved, long-term data on floating seaweed decomposition. They conducted an *ex-situ* study on the estuarine egg and bladder wracks *Ascophyllum nodosum* and *Fucus vesiculosus*, which both have pneumatocysts to stay afloat. Vandendriessche et al. (2007) exposed detritus of both species to four temperatures and grazing by the isopod *Idotea balthica*. The data show the broadest spectrum of decomposition trajectories in the published literature, so Vandendriessche et al. (2007) opted for non-parametric local regression to fit trendlines. Naturally this method is not inferential, so they resorted to pre-calculating longevity (days) and linear decomposition rates (%

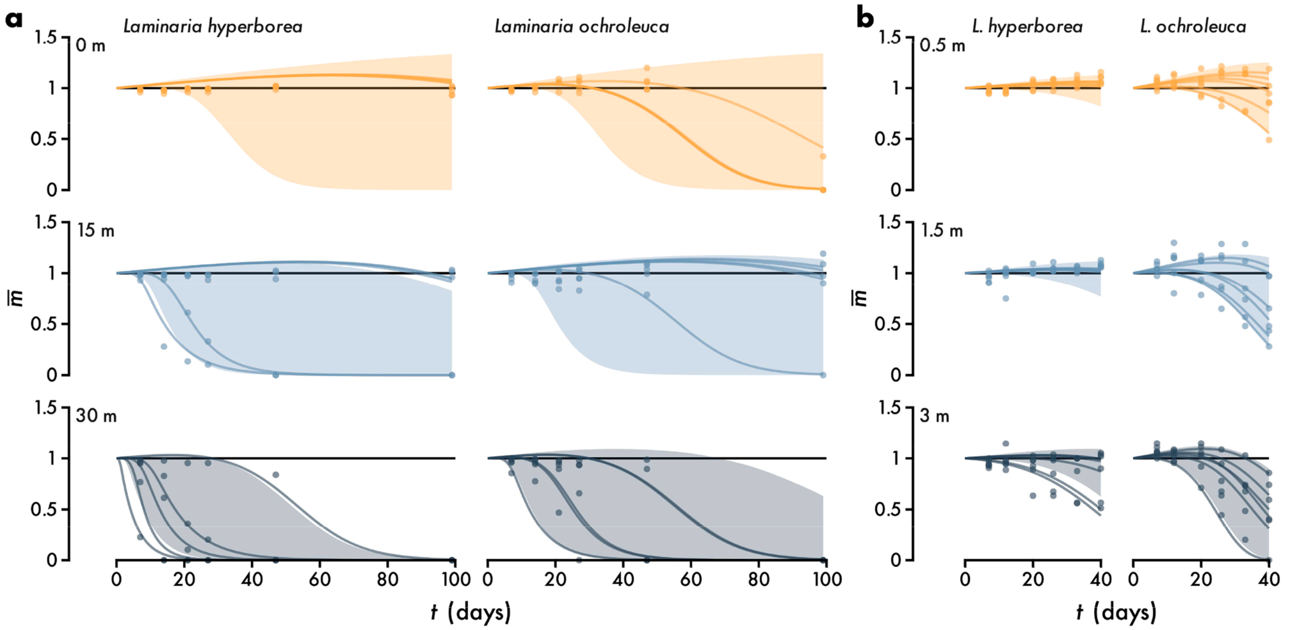

Fig. 5. Example 4: the effect of light attenuation with depth on the decomposition of *Laminaria hyperborea* and *Laminaria ochroleuca ex situ* (a, $n = 210$, data from Frontier et al. 2021) and in situ (b, $n = 252$, data from Frontier et al. 2022), shown as normalised remaining mass over time t. Data are plotted alongside predictions of the mean $\overline{m}$ where lines are medians per replicate and intervals are the central 90% of posterior probability across replicates (Eq. 19, cf. Fig. S5).

day^{-1}) per sample and reporting analysis of variance F statistics and Spearman's rank correlation coefficients for treatment effects. As with the previous example, replicates were repeatedly measured. Since a multilevel solution to pseudo-replication was not feasible in this case, likely due to highly divergent replicates, I could not partially pool replicates but included them with no pooling as

$$m_i \sim \mathrm{Beta}'(\,\overline{m}_i(1+\nu_i)\,,\,2+\nu_i\,) \tag{20}$$

$$\overline{m}_i = m(\,t_i\,,\,\alpha_i\,,\,\mu_i\,,\,\tau_i\,)$$

$$\nu_i = \nu(\,t_i\,,\,\epsilon\,,\,\lambda\,,\,\theta\,)$$

$$\ln\alpha_i = \ln\alpha_{S_i} + \ln\alpha_{R_i}$$

$$\ln\mu_i = \ln\mu_{S_i} + \ln\mu_{T_i} + \ln\mu_{R_i} + \beta_{T_i}G_i$$

$$\ln\tau_i = \ln\tau_{S_i} + \ln\tau_{T_i} + \ln\tau_{R_i} + \gamma_{T_i}G_i$$

$$\ln\alpha_S \sim \mathrm{Normal}(\,\ln 0.004\,,\,0.2\,)$$

$$\ln\mu_S \sim \mathrm{Normal}(\,\ln 30\,,\,0.2\,)$$

$$\ln\tau_S \sim \mathrm{Normal}(\,\ln 0.1\,,\,0.2\,)$$

$$\ln\alpha_R,\ \ln\mu_T,\ \ln\mu_R,\ \ln\tau_T,\ \ln\tau_R,\ \beta_T,\ \gamma_T \sim \mathrm{Normal}(\,0\,,\,0.2\,)$$

$$\epsilon,\ \lambda,\ \theta \sim \text{Priors as in Eq. 14}$$

where both μ and τ are influenced by temperature T and grazing G. Unlike depth, temperature may elicit a peaked photosynthetic response (Wright et al., 2024), so is here treated categorically. The binary grazing variable G is effect coded as -0.5 for control and $+0.5$ for grazed and stratified by temperature since isopod metabolism is also expected to be temperature-dependent. Exponentiating β and γ therefore yields the multiplicative grazing effects on μ and τ. To highlight the flexibility of the macroalgal model, predictions for each replicate are shown in Fig. 6 (cf. Fig. S6).

Averaged across replicates and treatments, *A. nodosum* has 0.058 ± 0.043 % day^{-1} faster initial detrital growth ($P_{\Delta\alpha>0} = 0.91$) and 9.5 ± 1.6 % day^{-1} slower final decomposition ($P_{\Delta\tau>0} = 1$) but a 36 ± 4.5 days shorter photosynthetic half-life ($P_{\Delta\mu>0} = 1$) than *F. vesiculosus*. Increasing temperature strongly reduces photosynthetic half-life across species (Fig. 6). There is a 100 % probability that photosynthetic half-life is 13 ± 1.5 times the average at 5°C but only 0.72 ± 0.074, 0.33 ± 0.036 and 0.39 ± 0.041 times the average at 10, 15 and 18°C respectively. In the absence of detrital photosynthesis, there is practically no accelerating effect of temperature on decomposition: the decay rate of detritus *sensu stricto* is 1 ± 0.15, 0.89 ± 0.098, 0.92 ± 0.1 and 1.1 ± 0.12 times the average at 5, 10, 15 and 18°C ($P_{\tau_T>1} = 0.47$, 0.14, 0.2 and 0.86). The negative grazing effect on photosynthetic half-life strengthens with increasing temperature: 5°C ($e^{\beta} = 0.96 \pm 0.12$, $P_{\beta<0} = 0.66$), 10°C ($e^{\beta} = 0.6 \pm 0.072$, $P_{\beta<0} = 1$), 15°C ($e^{\beta} = 0.29 \pm 0.037$, $P_{\beta<0} = 1$), 18°C ($e^{\beta} = 0.22 \pm 0.026$, $P_{\beta<0} = 1$). In contrast, grazing accelerates final decomposition evenly across temperatures: 5°C ($e^{\gamma} = 2 \pm 0.34$, $P_{\gamma>0} = 1$), 10°C ($e^{\gamma} = 3.6 \pm 0.47$, $P_{\gamma>0} = 1$), 15°C ($e^{\gamma} = 2.1 \pm 0.27$, $P_{\gamma>0} = 1$), 18°C ($e^{\gamma} = 1.4 \pm 0.18$, $P_{\gamma>0} = 0.99$). These results suggest that the accelerating temperature effect on seaweed decomposition (Filbee-Dexter et al., 2022; Frontier et al., 2022; Wright et al., 2024) acts almost exclusively through detrital photosynthesis rather than decomposer metabolism, which agrees with the conjecture of Wright et al. (2024). The conventional model (Fig. S6) underestimates decomposition by up to 3.5 orders of magnitude and is only reasonably accurate for the most photosynthetically impaired detritus exposed to grazing at high temperatures (Tables 1 and 2).

5. Conclusion

Macroalgal decomposition has neither been consistently nor reliably modelled. But a realistic model is the key to understanding macroalgal blue carbon. I have presented the first decomposition model tailored to macroalgae by taking into account the theory of detrital photosynthesis. This is achieved by treating the exponential decay rate as a function of time rather than a constant, a function which is informed by the latest understanding of detrital photosynthesis. My model only has three parameters which describe the initial relative growth rate (α), the photosynthetic half-life (μ), and the final exponential decay rate (τ) which is directly comparable to that of the classic model (k). I have shown that

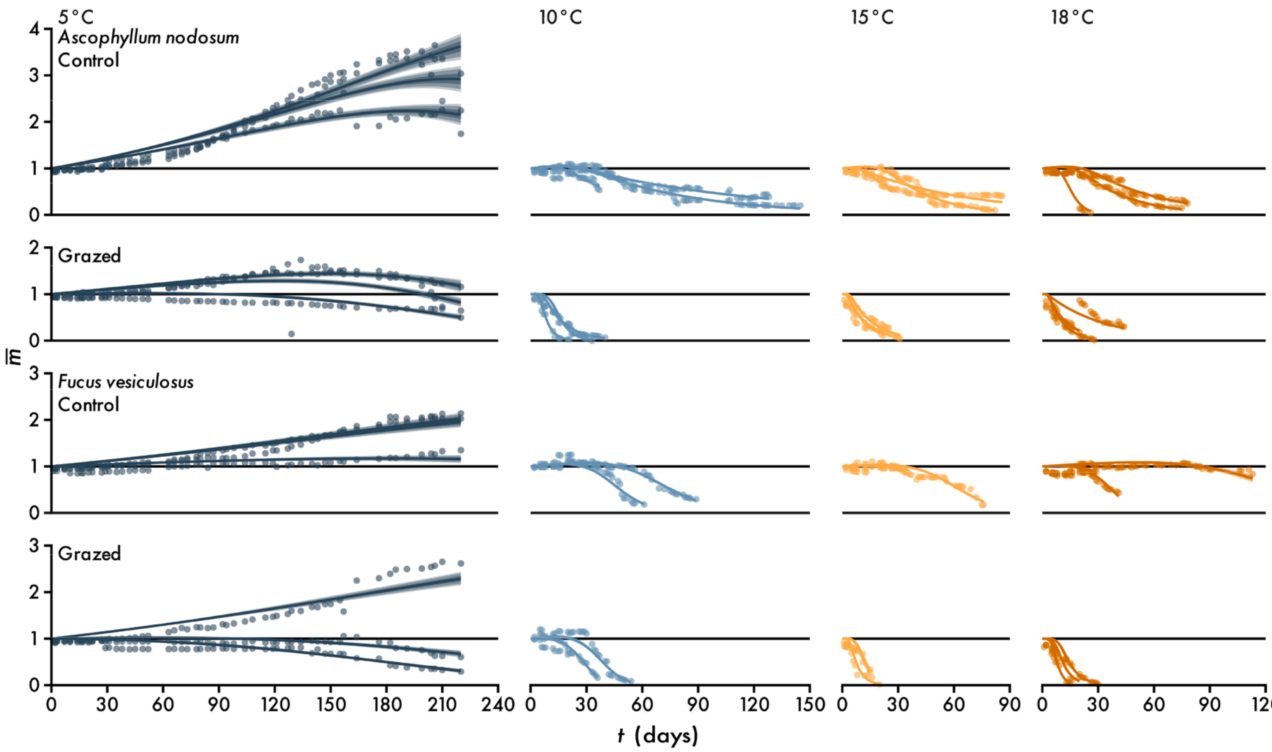

Fig. 6. Example 5: the effect of temperature and grazing on the decomposition of floating *Ascophyllum nodosum* and *Fucus vesiculosus* ($n = 1651$, data from Vandendriessche et al. 2007), shown as normalised remaining mass over time t. Data are plotted alongside predictions of the mean $\overline{m}$ for each replicate (Eq. 20, cf. Fig. S6). Lines and intervals are medians and the central 50, 80 and 90% of posterior probability.

these parameters are successfully estimated and ecologically interpretable for empirical data from a variety of contexts. Given a set of parameter estimates, the model can reliably predict new observations, even beyond the duration of the experiment. Although I have written the model in R with Stan and use a beta prime likelihood, it can be as effectively implemented with a gamma or even normal likelihood in any statistical framework that allows robust nonlinear modelling. Failing to account for detrital photosynthesis invariably leads to underestimation of macroalgal decomposition, especially when physiological influence is strong, leading me to believe that the conventional model is inadequate for these plants. I expect that the presented model will increase estimates of macroalgal decomposition in most cases and thus have major implications for seaweed blue carbon. I am already applying this model to my own kelp decomposition data and in meta-analysis and hope you may find it useful too.

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CRediT authorship contribution statement

Luka Seamus Wright: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

I declare no conflicts of interest. Importantly I am not funded by any private enterprise with an interest in selling carbon credits.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi 10.1016/j.ecolmodel.2026.111549.

Data availability

Data and code to reproduce this paper and apply my model to your own data are available at github.com/lukaseamus/limbodeco. Please cite this paper as well as the model (Wright 2026a).

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